



# Bruno Santos

Researcher in Statistics

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## Education

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| <b>PhD</b> | <b>University of Sao Paulo, Statistics</b> <ul style="list-style-type: none"> <li>Supervisor: Heleno Bolfarine</li> <li>Thesis title: Extensions of Bayesian quantile regression models</li> <li>Research visit at Duke University (2013-2014) under Alan Gelfand and Surya Tokdar</li> </ul> | Sao Paulo, Brazil<br>Jan 2012 – Dec 2016 |
| <b>MSc</b> | <b>University of Sao Paulo, Statistics</b> <ul style="list-style-type: none"> <li>Supervisor: Silvia Nagib Elia</li> <li>Dissertation title: Quantile regression models</li> </ul>                                                                                                            | Sao Paulo, Brazil<br>Jan 2010 – Dec 2012 |
| <b>BSc</b> | <b>University of Sao Paulo, Statistics</b>                                                                                                                                                                                                                                                    | Sao Paulo, Brazil<br>Jan 2005 – Dec 2008 |

## Experience

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| <b>CEAUL (Centro de Estatística e Aplicações - Universidade de Lisboa)</b> , Researcher in Statistics                                                                                                                                                      | Lisbon, Portugal<br>Apr 2024 – Dec 2025<br>1 year 9 months             |
| <b>University of Kent</b> , Lecturer in Statistics <ul style="list-style-type: none"> <li>Teaching: Statistics I; Probability and Classical Inference (2021/2022); Advanced Statistical Modelling; Predictive Modelling (2022/2023).</li> </ul>            | Canterbury, United Kingdom<br>Sept 2021 – Dec 2023<br>2 years 4 months |
| <b>Federal University of Espírito Santo (UFES)</b> , Assistant Professor <ul style="list-style-type: none"> <li>Teaching: Statistics I; Statistics II; Probability and Statistics I; Analysis of Categorical Data; Computational Statistics II.</li> </ul> | Vitoria, Brazil<br>Oct 2019 – July 2021<br>1 year 10 months            |
| <b>Georg-August-Universität Göttingen</b> , Postdoctoral Researcher <ul style="list-style-type: none"> <li>Teaching: Generalized Regression; Spatial Statistics.</li> </ul>                                                                                | Göttingen, Germany<br>Feb 2018 – Sept 2019<br>1 year 8 months          |
| <b>Federal University of Bahia (UFBA)</b> , Assistant Professor <ul style="list-style-type: none"> <li>Teaching: Statistical Methods; Bayesian Inference.</li> </ul>                                                                                       | Salvador, Brazil<br>Dec 2016 – Jan 2018<br>1 year 2 months             |

## Publications

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| <b>A defective cure rate quantile regression model for male breast cancer data</b><br>Rodrigues, A., Borges, P., Santos, B.<br>(Journal of Applied Statistics, 52(8), 1485-1512) | 2025 |
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| <b>Using population register data and capture-recapture models to estimate over-coverage in Sweden</b>                                                                 | 2024 |
| Santos, B., Mussino, E., Drefahl, S., Matechou, E.<br>(Scientific Reports 14, 30551)                                                                                   |      |
| <b>Multiple systems estimation for studying over-coverage and its heterogeneity in population registers</b>                                                            | 2024 |
| Mussino, E., Santos, B., Monti, A., Matechou, E., Drefahl, S.<br>(Quality & Quantity 58, 5033-5056)                                                                    |      |
| <b>Bias reduction of maximum likelihood estimates for an asymmetric class of power models with applications</b>                                                        | 2023 |
| Gomez, Y., Santos, B., Gallardo, D., Venegas, O., Gomez, H.<br>(REVSTAT - Statistical Journal, 21(4), 491-507)                                                         |      |
| <b>Noncrossing structured additive multiple-output Bayesian quantile regression models</b>                                                                             | 2020 |
| Santos, B., Kneib, T.<br>(Statistics and Computing)                                                                                                                    |      |
| <b>Bayesian quantile regression analysis for continuous data with a discrete component at zero</b>                                                                     | 2018 |
| Santos, B., Bolfarine, H.<br>(Statistical Modelling)                                                                                                                   |      |
| <b>Bayesian analysis for zero-or-one inflated proportion data using quantile regression</b>                                                                            | 2015 |
| Santos, B., Bolfarine, H.<br>(Journal of Statistical Computation and Simulation)                                                                                       |      |
| <b>Influence measures in quantile regression models</b>                                                                                                                | 2015 |
| Santos, B., Elian, S.<br>(Communications in Statistics - Theory and Methods)                                                                                           |      |
| <b>Association of pollution with quantiles and expectations of the hospitalization rate of elderly people by respiratory diseases in the city of Sao Paulo, Brazil</b> | 2014 |
| Alencar, A., Santos, B.<br>(Environmetrics)                                                                                                                            |      |

## Conference Proceedings

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| <b>Growth curves for multiple-output response variables via Bayesian quantile regression models</b>               | 2020 |
| Santos, B., Rodrigues, A., Kneib, T.<br>(Proceedings of the 35th International Workshop on Statistical Modelling) |      |
| <b>Structured additive multiple-output noncrossing Bayesian quantile regression models</b>                        | 2018 |
| Santos, B., Kneib, T.<br>(Proceedings of the 33rd International Workshop on Statistical Modelling)                |      |
| <b>Analysis of Brazil's presidential election via Bayesian spatial quantile regression</b>                        | 2015 |
| Santos, B., Bolfarine, H.<br>(Proceedings of the 30th International Workshop on Statistical Modelling)            |      |
| <b>A two-part model using quantile regression under a Bayesian perspective</b>                                    | 2013 |
| Santos, B., Bolfarine, H.<br>(Proceedings of the 28th International Workshop on Statistical Modelling)            |      |

**Analysis of residuals in quantile regression: an application to income data in Brazil** 2012  
Santos, B., Elia, S.  
(Proceedings of the 27th International Workshop on Statistical Modelling)

## Awards

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**Collaborations with Developing Countries, Grant Scheme 5** 2022  
London Mathematical Society (Grant amount: GBP 1,972)

**2nd place, Best poster award** 2018  
23rd Brazilian National Symposium of Probability and Statistics

**Best Student Poster Toyota Corporation Award** 2015  
30th International Workshop on Statistical Modelling (USD 500)

**Student Travel Grant** 2013  
28th International Workshop on Statistical Modelling (EUR 800)

## Teaching

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University of Kent (2021/2022): Statistics I; Probability and Classical Inference  
University of Kent (2022/2023): Advanced Statistical Modelling; Predictive Modelling  
Federal University of Espirito Santo (2019-2021): Statistics I; Statistics II; Probability and Statistics I; Analysis of Categorical Data; Computational Statistics II  
Georg-August-Universität Göttingen (2018-2019): Generalized Regression; Spatial Statistics  
Federal University of Bahia (2017): Statistical Methods; Bayesian Inference

## Short Courses

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3rd Paraíba Meeting of Statistics (2021): Regression models beyond the conditional mean  
24th Brazilian National Symposium of Probability and Statistics (postponed, 2020): Visualization of spatial effects in regression models  
XIII School of Regression Models (2013): Regression models with truncated and censored responses  
Universidad de Antofagasta (2017): Quantile regression: a classical and a Bayesian approach

## Service and Activities

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Member, Migrating Trajectories research project (University of Stockholm and University of Kent)  
Local organizing committee, Movement and migration in ecology workshop (University of Kent, May 2022)  
Local organizing committee, 1st Workshop on Data Science and Statistical Learning (UFES, November 2019)  
Leader, Observatory of Covid-19 in favelas of Espirito Santo (OCOFA)  
Program Committee member, useR! 2021 and useR! 2022 conferences

## Software

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### **baquantreg**

R package for Bayesian quantile regression.  
• GitHub: <https://github.com/brsantos/baquantreg>

### **ggcrqr**

R package for quantile regression visualization and diagnostics.  
• GitHub: <https://github.com/brsantos/ggcrqr>